

PrimeFocus

February 2026



WHEN

February 20, 2026

Doors open at 7:00pm

Meeting starts at 7:30pm

Lecture starts at 8:00pm

WHERE

Unitarian Church

1893 North Vasco Road

Livermore, CA 945571

and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

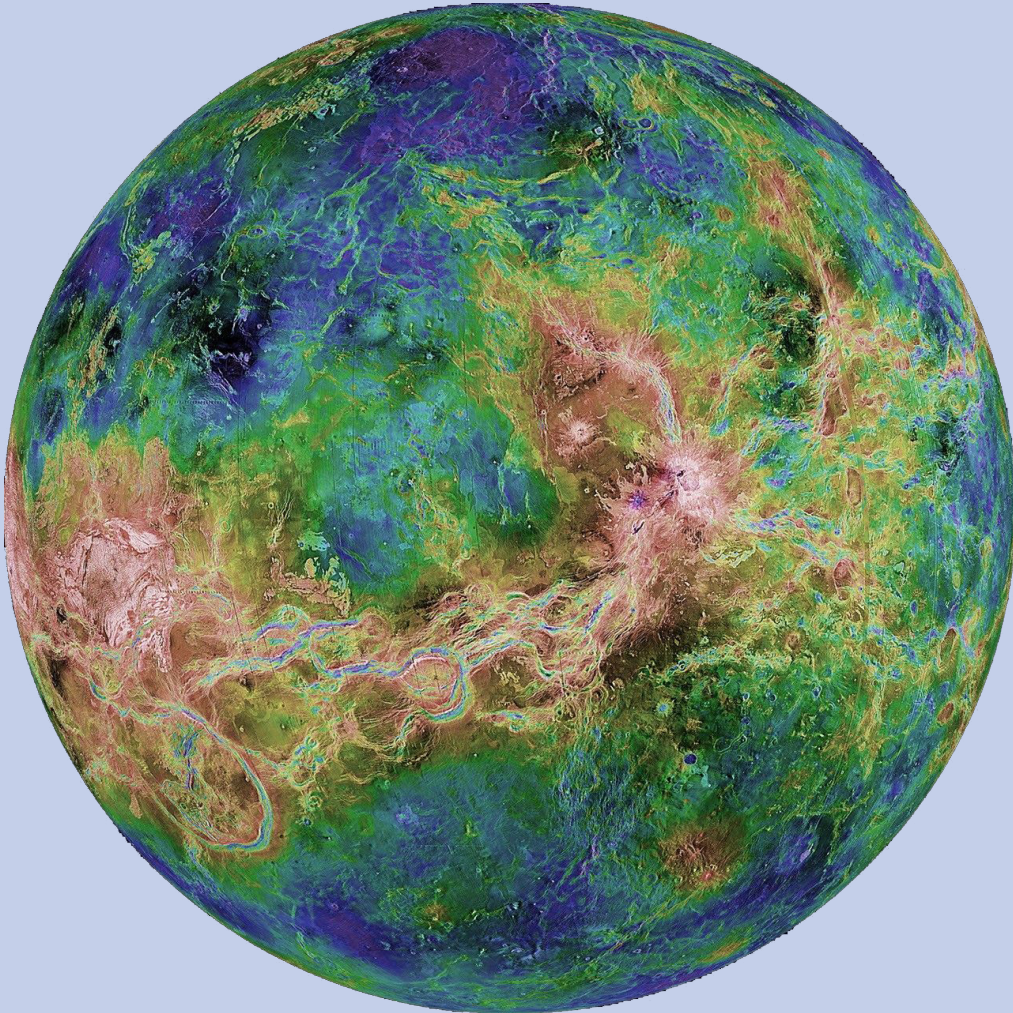
Venus as a Potentially Habitable Planet **Professor Sara Seager**

This month's presentation will be a recording of a talk presented to the Kalamazoo Astronomical Society on Friday, January 9, 2026.

Scientists have been speculating on Venus as a habitable world for over half a century, not the scorching surface, but the much cooler atmosphere at 48 to 60 km above the surface. The concept is that Venus' perpetual cloud cover might host life, as Earth's clouds do. The Venus clouds, however, are not made of water but are composed of concentrated sulfuric acid—an aggressive chemical that is toxic for Earth life. New lab-based experiments show some biomolecules are stable in concentrated sulfuric acid, and advance the notion that the Venus atmosphere environment may be able to support complex chemicals needed for life and motivate the astrobiology-focused Morning Star Missions to Venus.

About the Speaker:

Professor Sara Seager is a Canadian-American astrophysicist and Professor of Physics, Planetary Science, and Aeronautics and Astronautics at the Massachusetts Institute of Technology, where she holds the Class of 1941 Professor Chair. Her research ranges from the foundation of exoplanet atmospheres to innovative theories about life on other worlds to development of novel space mission concepts. She was Deputy Science Director of NASA's TESS mission, Principal Investigator of the JPL-MIT CubeSat ASTERIA, and has led major efforts to design space-based direct imaging missions to discover another Earth. She now leads the Morning Star Missions to Venus in search of signs of life in Venus' clouds. Her many honors include a MacArthur "genius" award, the Kavli Prize in Astrophysics, membership in the US National Academy of Sciences, and appointment as an Officer of the Order of Canada. She is the author of the textbook "Exoplanet Atmospheres" and also "The Smallest Lights in the Universe: A Memoir."



Radar Observations of Venus Image Credit: NASA/JPL-Caltech/USGS

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
February 20	February 23	February 7
March 20	March 23	March 7
April 17	April 20	April 4

Money Matters

As of the last Treasurer's Report on 12/22/25, our club's account balance is \$49,833.30, this includes \$12,569.51 in the H2O Rebuild fund. No treasury update during 1/19/26 board meeting.

TVS Welcomes New Members

TVS welcomes new members Pushparajan Vijayakumar, Sharlyn Dreyfus, and Nathan Woolsey. Please say hello and chat with them during our meetings.

2026 TVS Club Star Party Schedule

Save the dates for the 2025 Club Star Parties. Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

No events currently scheduled for Del Valle.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

No events currently scheduled for Tesla Vintners.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

No open house events currently scheduled for H2O.

February 26: LVJUSD Science Odyssey at Joe Mitchell TK-8, 1001 Elaine Ave, Livermore. from 5:00-7:00pm.

March 21: Public Star Party at Sunol Regional Wilderness, 1895 Geary Road, Sunol. Set-up 7:00pm, Observing from 7:45-8:45pm

CALENDAR OF EVENTS



February 20, 21, 27,28, March 6, 7, 13,14, 20, 21, 7:30-10:30pm

What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot's historic telescopes. Chabot's three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center's 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see <https://chabotspace.org/events/events-listing/> <https://eastbayastro.org/chabot-telescope-status/>

March 11, 2026, 7:00pm (Rescheduled from last month)

What	Pictures of Distant Worlds
Who	Silicon Valley Astronomy Lecture Series
Where	Smithwick Theater at Foothill College
Cost	Free

In the past three decades, more than 6000 planets have been discovered orbiting other stars beyond our own Solar System. However, we still don't know if our Solar System is rare or unique — the powerful techniques that detect extrasolar planets have discovered systems very different than our own. In recent years, advances in technology have allowed a handful of giant planets around other stars to be imaged directly. Find out about the first-ever images of other solar systems — and the technology that has allowed us to discover them, such as the Gemini Planet Imager — as well as the future planet-hunting space telescopes. The ultimate goal is detection of a second 'pale blue dot' —

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an Earth twin where we could even see the biosignatures of extrasolar life.



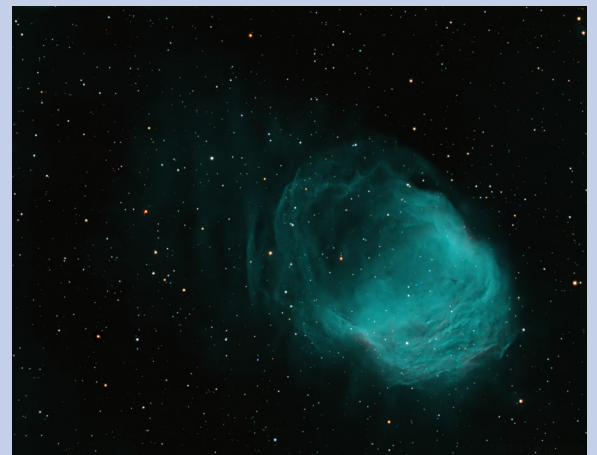
Gemini Planet Imager (GPI) on Gemini South telescope. Credit: International Gemini Observatory/NOIRLab/NSF/AURA/M. Paredes

Bruce Macintosh is the Director of the University of California Observatories in California and Hawaii. His research focuses on the study of extrasolar planets, in particular the study of such planets through direct imaging. Direct imaging of extrasolar planets involves blocking, suppressing, and subtracting the light of the bright parent star so that a planet hundreds of thousands of times fainter can be seen and studied in detail. Dr. Macintosh co-led the team that imaged the first extrasolar planets, and was the Principal Investigator of the Gemini Planet Imager, an advanced adaptive optics planet-finder for the Gemini South telescope.



For parking information, see <https://foothill.edu/parking/>





The Medusa Nebula, Abell 21, by Scott Schneider. Top image is a combination of RGB, and SII, Ha, and OIII. Bottom left is RGB and Ha. Bottom right image is RGB and OIII. For a full resolution image please see <https://app.astrobin.com/i/n7y55c>

WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

February 2026

23	Mon	Looking southwest at dusk see the moon next to the Pleiades
24	Tue	Moon is at first quarter
27	Fri	In the morning see the moon with Jupiter 3° lower left. Then at dusk see the min form a line with Caster and Pollux. The moon is just 4° below Pollux, and Jupiter is father to the right

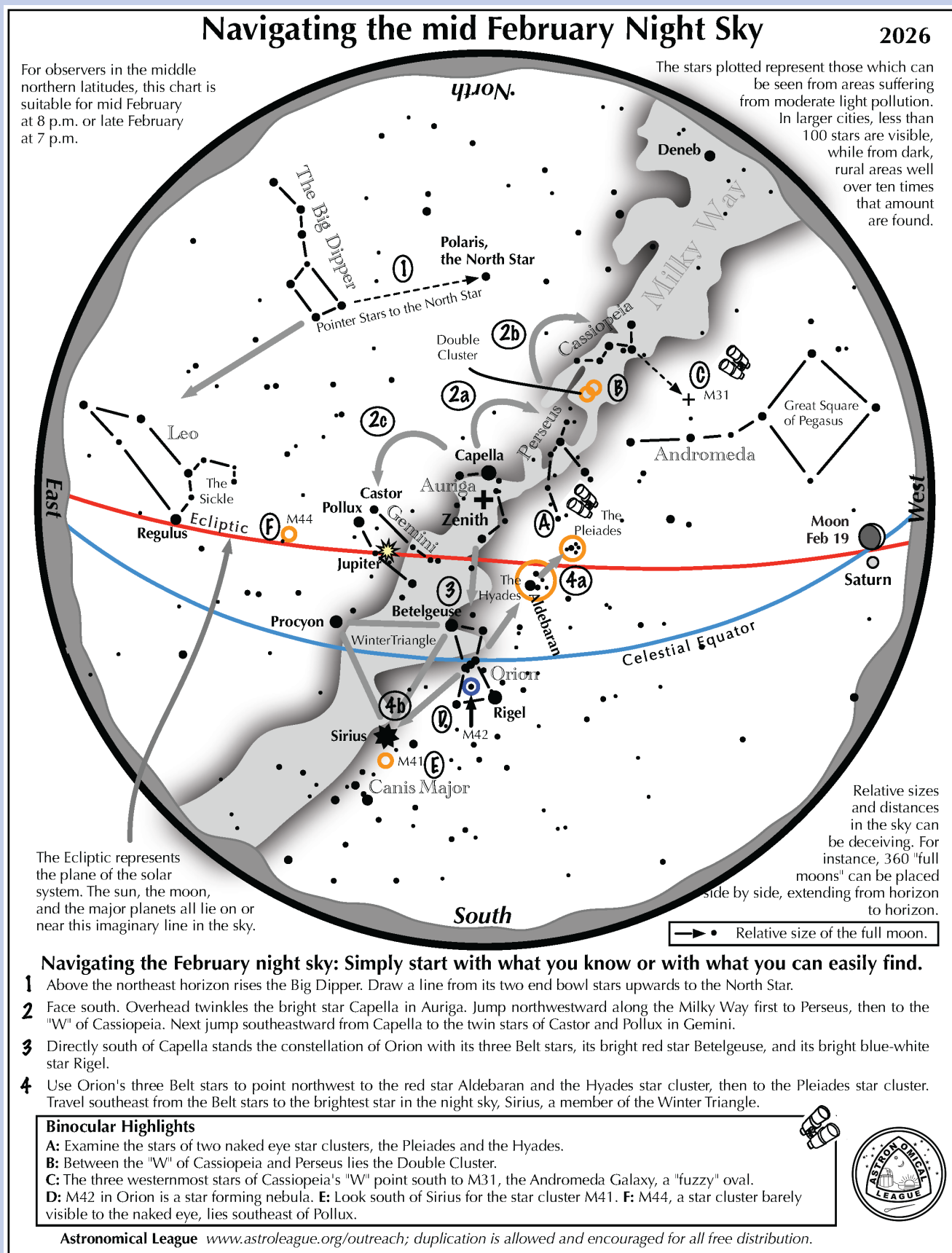
March 2026

2	Mon	Watch the moon rise in the east with Regulus, As the pair rise the moon occults the star.
3	Tue	Full Moon
6	Fri	In the morning, looking southeast, Moon is just 3½° upper right of Spica
8	Sun	Daylight savings time begins at 2:00 AM
10	Tue	At dawn, moon is just 1½° lower right of Antares
11	Wed	Moon is at last quarter
13	Fri	Algol shines at minimum brightness between 8:07 and 10:07 PM
19	Thu	New Moon

OFFICERS AND VOLUNTEER POSITIONS

Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
Vice-President Aris Pope vice_president@trivalleystargazers.org	Club Star Party Coordinator Johnathan Bailey coordinator@trivalleystargazers.org	Webmaster Swaroop Shere webmaster@trivalleystargazers.org	Program Coordinator Ron Kane programs@trivalleystargazers.org
Treasurer Ernest Epley treasurer@trivalleystargazers.org	Del Valle Coordinator David Wright delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H2O@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Johnathan Bailey outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR FEBRUARY 2026





Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.